Chapter 14 Oscillations

This chapter studies periodic motion of objects such as springs and pendulums. Oscillations appear throughout physics and engineering.

14.1 Simple Harmonic Motion

When a mass is attached to an ideal spring, the spring exerts a restoring force.
Hooke’s law states that the restoring force is proportional to displacement, F = -kx.
The resulting motion is called simple harmonic motion.
The angular frequency of the mass-spring system is omega = sqrt(k/m).
The period of oscillation is T = 2 pi sqrt(m/k) and does not depend on amplitude.
The amplitude is the maximum displacement from the equilibrium position.
Determine the angular frequency of a mass-spring system if k = 200 N/m.

14.2 Energy in Simple Harmonic Motion

In simple harmonic motion kinetic energy and potential energy continuously exchange.
The total mechanical energy of the oscillator is $E = 1/2\ k\ A^2$.
Because there is no friction, the total energy remains constant.

14.3 The Simple Pendulum

A simple pendulum consists of a point mass suspended from a light string of length L.
For small angles the motion is approximately simple harmonic.
The period of a simple pendulum for small angles is T = 2 pi sqrt(L/g).

14.4 Damped Oscillations

Real oscillators lose energy because of friction or air resistance.
A damping force proportional to velocity causes the amplitude to decrease over time.
Motion can be underdamped, critically damped or overdamped depending on the damping.

Chapter 15 Mechanical Waves

A wave is a disturbance that travels through a medium.
Transverse and longitudinal waves differ in the direction of displacement.